# Analysis Tutorial 4

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## 1 Introduction

In this tutorial, we will explore equivalence relations, equivalence classes, and some of their applications to integers, rational numbers, and graphs.

### Equivalence Relation

A relation $\sim$ on a set $X$ is an *equivalence relation* if it satisfies the following properties:

- Reflexive: $\forall x \in X,\ x \sim x$,
- Symmetric: $\forall x, y \in X,\ (x \sim y \implies y \sim x)$,
- Transitive: $\forall x, y, z \in X,\ (x \sim y \wedge y \sim z \implies x \sim z)$.

### Equivalence Class

Given $x \in X$, the *equivalence class* of $x$ is the set

$$[x] = \{y \in X : x \sim y\}.$$

In other words, $[x]$ is the set of all elements of $X$ that are equivalent to $x$. Remark: Equivalence classes are disjoint (they do not share elements).

### Well-definedness

When defining functions $f : X/\sim \to Y$, issues may arise regarding whether $f$ is well-defined.

**Example.** Define $f : \mathbb{Z}/5\mathbb{Z} \to \mathbb{Z}$ by $f([m]) = m$. Is this well-defined?

## Problems:

**Problem 1.** Let $S$ be a set with $|S| = n$. Prove that $|\mathcal{P}(S)| = 2^n$, where $\mathcal{P}(S)$ denotes the power set of $S$.

**Problem 2.** Let $X = Y = \mathbb{Z}$. Define a relation $R$ on $X$ by

$$R(a, b) \iff a \mid b.$$

Is $R$ an equivalence relation? If not, how can we modify it to make $R$ an equivalence relation?

**Problem 3.** Let $G = (V, E)$ be an undirected graph. Define a relation $\sim$ on $V$ by

$$u \sim v \iff \text{there exists a path in } G \text{ between } u \text{ and } v.$$

(a) Prove that $\sim$ is an equivalence relation.
(b) Describe the equivalence classes of $\sim$.

**Problem 4.** Define a relation $\equiv$ on $V$ by

$$u \equiv v \iff \deg(u) = \deg(v).$$

(a) Determine whether $\equiv$ is an equivalence relation.
(b) For the cycle graph $C_5$, compute the equivalence classes.

**Problem 5.** Prove that there is no bijection between $[m]$ and $[n]$ where $m$ and $n$ are distinct positive integers.

**Problem 6.** One can also define an equivalence relation by declaring two sets equivalent if there exists a bijection between them. List all possible types of equivalence classes that arise from this definition.

**Problem 7.** Construct a bijection between $[0, 1)$ and $[0, \infty)$.

**Problem 8.** Prove that there exists a bijection between the unit circle $S^1$ and the interval $[0, 1)$.

**Problem 9.** Let $\mathbb{N}$ denote the set of natural numbers (for definiteness one may take $\mathbb{N} = \{0, 1, 2, \dots\}$).

**Problem A.** Construct a bijection

$$f : \mathbb{N} \longrightarrow \mathbb{N}^2.$$

**Problem B.** For each integer $k \geq 1$, construct a bijection

$$g_k : \mathbb{N} \longrightarrow \mathbb{N}^k.$$